1. Wu et al. Diurnal and Seasonal Variations of Meteor Velocity Observed at the Middle Latitude by the Mengcheng Meteor Radar. Submitted to *JGR Space Phys*; oral at *20th Nat Symp on Solar-Terrestrial Space Phys*

Key Points • Azimuthal direction of the maximum meteor velocity changes clockwise, from the north (local morning) to the south (local evening) • Meteor velocity distribution has two Gaussian peaks at ~28 km/s and ~54 km/s, revealing bi-annual and annual variations respectively • Crucial meteor showers occur with 45–97% higher counts than the background, causing velocity peaks to increase

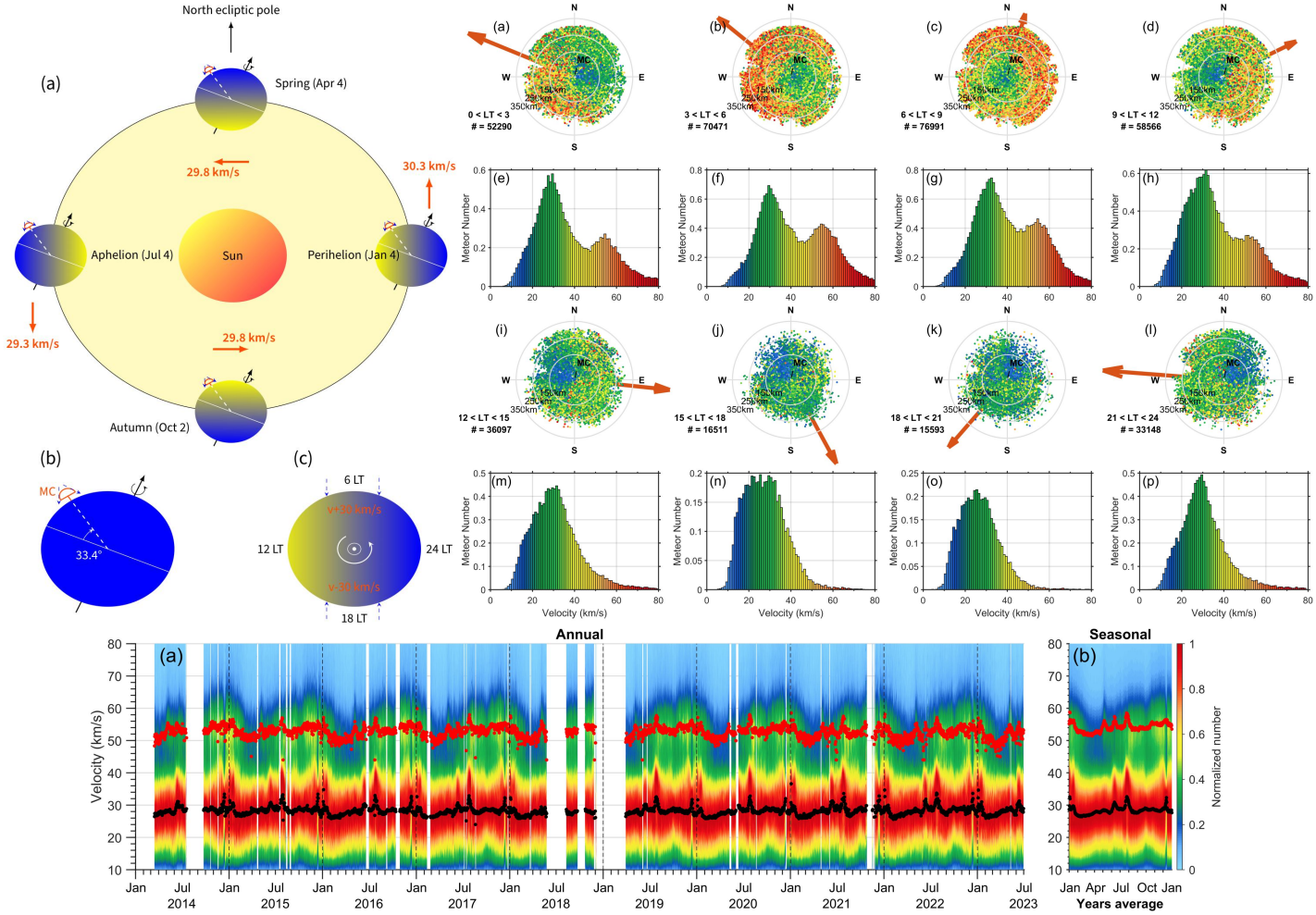


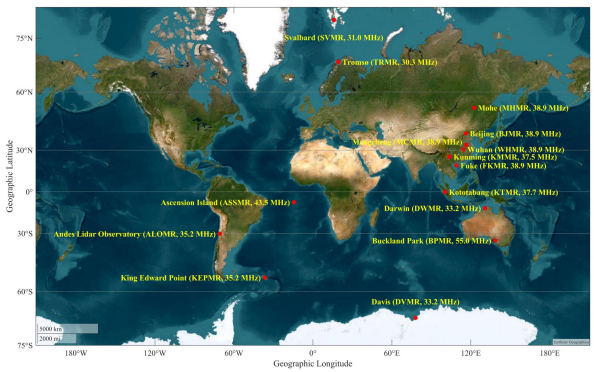
Fig. 3 of Pub. 1 (top-left). Schematic illustrating the viewing geometry of the MCMR. (a) the position of MCMR in the solar system changes as a result of Earth's rotation and Earth's orbit around the Sun; (b) the location of MCMR on Earth; (c) the viewing geometry of MCMR varies with local time, causing variation in the observed meteor velocity and angle of arrival, see down from the North Ecliptic Pole. **Fig. 7 of Pub. 1** (top-right). The first and third rows depict distribution maps of meteor velocities as a function of horizontal distance and azimuth. These are sorted into 8 local time intervals over the composition of a 30-day period centered on October 2, 2022. Each meteor is represented as a colored point, with different colors used to indicate its velocity, as shown in the second and fourth rows where color scales are provided. Notably, " LT" is an abbreviation for Mengcheng local time (approximately UT+8 h). The circles in maps in the first and third rows denote distances of 150 km, 250 km, and 350 km. **Fig. 9 of Pub. 1** (bottom). Panel (a) shows the nine-year variations of daily meteor velocity distributions, in which red and black dots each denote the daily lower velocity peak (μ_1) and the higher velocity peak (μ_2) observed by MCMR from April 2014 to June 2023. The color bar denotes the daily normalized number density. Panel (b) is the composite average of nine-year velocity distributions for Panel (a).

2. Yi et al. (including Wu, K.) Climatology of the Global Mesospheric Diurnal Tides Observed by Meteor Radars. submitted to *Geophys Res Lett*

Key Points • The climatology of the global diurnal tide was observed by fifteen meteor radars in the MLT region. • The diurnal tides are strong at low and mid-latitudes (10-50°N/S) and are weak at the equator, higher-mid and polar latitudes. • The height of the maximum tidal amplitude is related to the solar zenith angle at northern low and mid-latitudes.

3. Wu et al. A Summary Report on the Space Physics Practical Education in 2022. published at *Rev Geophys and Planetary Phys*

Keywords: practical education; Space Physics; Chinese Meridian Project



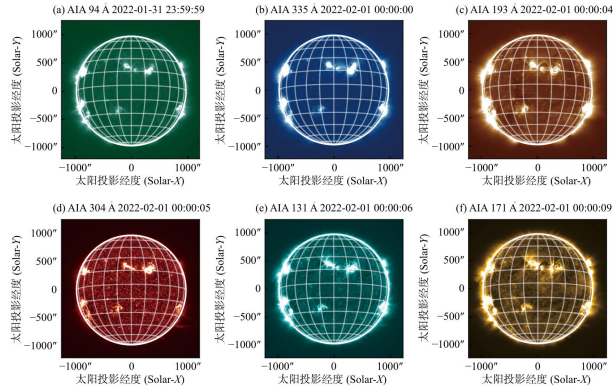


Fig. 1 of Pub. 2 (top-left). The locations and working frequencies of the meteor radars. **Fig. 3 of Pub. 3** (top-right). Images of solar disks at different bands observed by the SDO satellite from 23:59:59 UT on January 31 to 00:00:09 UT on February 1, 2022. Panels (a) to (f) show the images in the 94 Å, 335 Å, 193 Å, 304 Å, 131 Å and 171 Å bands.

4. Wu et al. Perturbations by the 2022 Hunga-Tonga Volcano Eruption in the Mesosphere-Lower Thermosphere Region Investigated Using the WACCM-X Simulation and Meteor Radar Observations. oral at *4th CGS Workshop*; poster at *AGU 2023;* prepare to submit to *Geophys Res Lett*

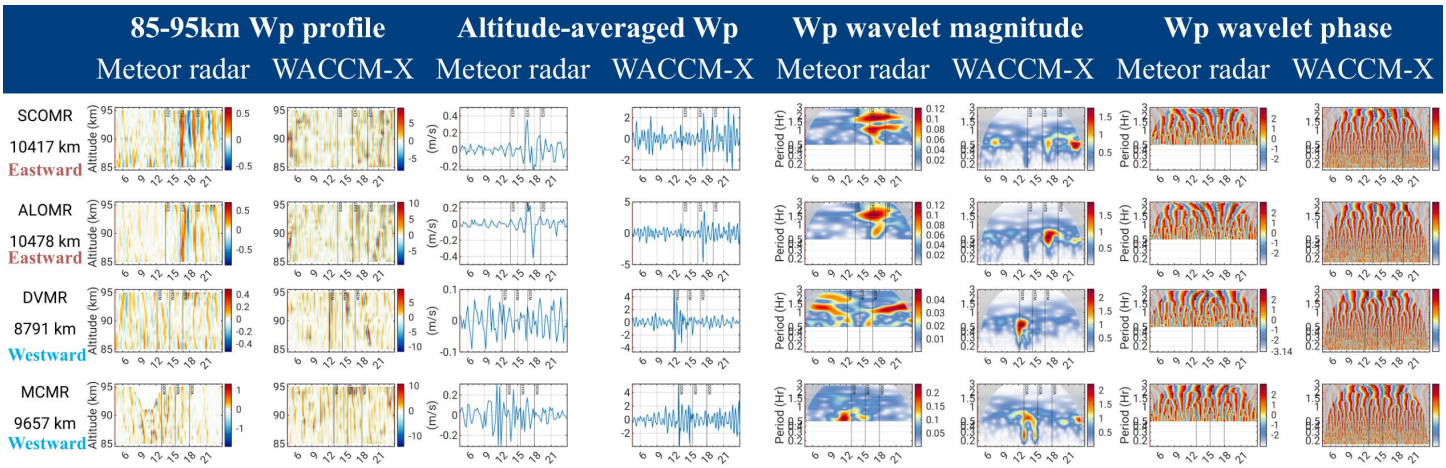


Fig. 1 of Pub. 4. Analysis of 1-hour high-pass filtered Wp (Horizontal wind vectors projected towards the great circle direction from the epicenter to the meteor radar stations).

5. 6. 7. 8. Wu et al. Other presentations (including first-author presentations and invited presentations) & hobby

(From left to right:) **April '23** *20th National Symposium on Solar-Terrestrial Space Physics* (The only undergraduate among more than 100 scholars in the mid-upper atmosphere branch), **May '23** *Top-notch Innovative Talent Joint Training Symposium* (The only undergraduate invited among more than 40 scholars), **June '23** *USTC Lecture* (The only undergraduate invited to present among 150+ attendees), **November '23** *4th CGS (NASA DRIVE Science Center for Geospace Storms) Workshop*, **March '23** rock climbing.